

AI Job Market Analysis

A comprehensive Power BI dashboard analyzing AI job demand, salaries, experience levels, and work types.

Presented by: Haseema Ibrahimbavage

Project Scope & Objectives



Track Job Openings

Track AI vacancies



Analyze Salary Trends

Analyze compensation patterns



Evaluate Work Models

Assess work arrangements



Support Decisions

Provide data-driven insights



Identify Skill Requirements

Determine key AI skills



Track Company Hiring Patterns

Monitor recruitment trends

INTRODUCTION : Navigating the AI Frontier



Market Expansion

Rapid growth in the AI job market due to technological advancements.



Power BI Analysis

Power BI analyzes complex data for clear insights.



Actionable Insights

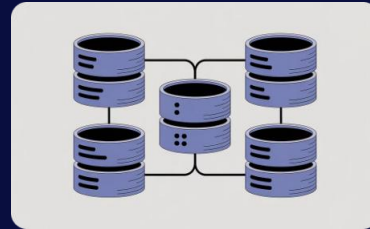
Provide actionable insights to help AI stakeholders make informed decisions.

Methodology: Power BI Dashboard Development



1. Data Import & Preparation

Import, clean, and transform data.



2. Data Modeling

Model data, create calculated columns.



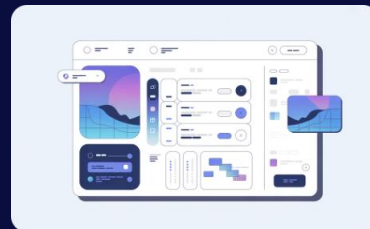
3. DAX Formulas for KPIs

Develop DAX for key KPIs.



4. Visualization Design

Design interactive charts and KPIs.



5. Interactive Slicers

Implement interactive filters.



6. Dashboard Layout

Design two-page dashboard layout.

Dataset

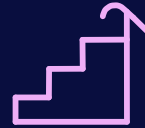
	A	B	C	D	E	F	G	H	I	J	K	L
1	job_id	company_name	industry	job_title	skills_required	experience_level	employment_type	location	salary_range_usd	posted_date	company_size	tools_utilized
2	1	Footer and Sons	Healthcare	Data Analyst	NumPy, Reinforcement Learning, PyTorch, Solidi-learn, GCP, FastAPI	Mid	Full-time	Tracybury, AR	92860-100688	20-08-2025	Large	KCB*, LangChain
3	2	Gord. Myers and Ramirez	Tech	Computer Vision Engineer	Solidi-learn, CUDA, SQL, Pandas	Senior	Full-time	Lake Scott, CU	72523-144875	23-09-2024	Large	FastAPI, KDB*,
4	3	King Inc	Tech	Quant Researcher	MLflow, FastAPI, Azure, PyTorch, SQL, GCP	Entry	Full-time	East Paige, CM	124498-217804	18-08-2028	Large	BigQuery, PyTorch, Solidi-learn
5	4	Conper, Archer and Lynch	Tech	AI Product Manager	Solidi-learn, C++, Pandas, LangChain, AWS, R	Mid	Full-time	Pereclev, FI	60908-129743	09-08-2024	Large	TensorFlow, BigQuery, MLflow
6	5	Hall LLC	Finance	Data Scientist	Excel, Keras, SQL, Hugging Face	Senior	Contract	Noah Oesteland, NE	98894-185419	24-02-2026	Large	PyTorch, LangChain
7	6	Ellis PLC	E-commerce	AI Product Manager	GCP, Excel, Solidi-learn, MLflow	Senior	Remote	South Keim, TZ	92892-180718	07-08-2026	Large	PyTorch, TensorFlow,
8	7	Menthaue-Moses	Automotive	Data Analyst	Python, Flask, Solidi-learn, GCP	Mid	Full-time	West Shawn, LA	70679-80594	23-12-2024	Large	PyTorch, MLflow
9	8	Mullins Ltd	Education	Data Scientist	Excel, SQL, Flask	Entry	Internship	Port Harlow, RU	68032-82144	14-08-2024	Large	BigQuery
10	9	Asulka PLC	Healthcare	ML Engineer	Pandas, LangChain, GCP, Azure	Entry	Internship	Budermouth, GB	194239-188804	18-09-2026	Large	Solidi-learn, LangChain,
11	10	Parish LLC	Automotive	Computer Vision Engineer	Excel, Reinforcement Learning, TensorFlow	Senior	Full-time	Nooleshire, BA	79361-166029	07-08-2026	Large	MLflow, TensorFlow
12	11	Venaghan-Orie	Tech	Data Scientist	Keras, MLflow, Reinforcement	Entry	Internship	New Angelatown, LU	39384-128871	10-01-2024	Large	LangChain, BigQuery,
13	12	Johnson, Guzman and Hernandez	Finance	ML Engineer	Excel, Reinforcement Learning, Keras, Solidi-learn	Entry	Full-time	Joychester, LT	63889-193870	15-03-2024	Large	TensorFlow, PyTorch, BigQuery
14	13	Martin and Sons	E-commerce	NLP Engineer	C++, Hugging Face, Python, FastAPI	Entry	Full-time	South Michael, FM	102770-111020	27-09-2023	Startup	LangChain
15	14	Frank Osane	Retail	Quant Researcher	Excel, MLflow, NumPy, Hugging Face, LangChain, FastAPI	Senior	Full-time	Christenberg, GQ	137004-188825	20-11-2024	Mid	LangChain, TensorFlow, Solidi-learn
16	15	Rogert Ltd	Education	NLP Engineer	Pandas, CUDA, NumPy, FastAPI	Mid	Remote	New Katieborough, FJ	128342-216741	08-07-2024	Large	FastAPI
17	16	Wright Osane	Automotive	NLP Engineer	Hugging Face, Solidi-learn, Reinforcement Learning	Senior	Remote	Stephanlechester, LU	128889-217888	09-08-2024	Startup	TensorFlow
18	17	Preston Ltd	E-commerce	ML Engineer	MLflow, R, C++	Entry	Full-time	South Tim, C2	137523-180097	14-06-2026	Mid	LangChain
19	18	Smith-Moore	Automotive	AI Product Manager	Excel, MLflow, HuggingFace, Flask, Power BI	Entry	Internship	Blevineburgh, VU	133184-182657	29-04-2024	Startup	BigQuery
20	19	Webb and Sons	Healthcare	Data Scientist	AWS, Azure, Power BI, CUDA, Reinforcement Learning, Hugging	Senior	Remote	Portsmouth, CA	100846-188745	19-10-2024	Mid	BigQuery, FastAPI, KDB*
21	20	Gates LLC	E-commerce	ML Engineer	R, CUDA, Power BI, AWS	Mid	Internship	Steenhacen, BW	116578-207180	28-08-2024	Large	MLflow
22	21	Wistrater, Walker and Sandcoal	Retail	Quant Researcher	C++, Power BI, R, SQL, Keras, Azure	Entry	Remote	Gregorymouth, LV	122521-175188	18-01-2024	Large	Solidi-learn, BigQuery, PyTorch
23	22	Humphrey LLC	Retail	Quant Researcher	LangChain, NumPy, CUDA, C++, MLflow, R	Entry	Full-time	Lake Alexander, RU	62489-106216	26-12-2023	Large	PyTorch, LangChain
24	23	Adams, Stevenson and Wegner	E-commerce	NLP Engineer	CUDA, FastAPI, Solidi-learn, MLflow, Hugging Face	Mid	Remote	North Nicholas, SN	74317-148899	26-11-2024	Startup	PyTorch, Solidi-learn, KCB*
25	24	Thames, Hunt and Becker	Tech	NLP Engineer	PyTorch, CUDA, Hugging Face, Flask, C++	Senior	Internship	Leaton, MW	88116-126328	08-12-2023	Mid	Solidi-learn
26	25	Sherman-Nguyen	Healthcare	NLP Engineer	Python, Reinforcement Learning, MLflow	Mid	Internship	Sucannectev, TW	88840-108633	18-08-2025	Mid	FastAPI
27	26	Johnson-Keller	Finance	NLP Engineer	PyTorch, Reinforcement Learning, FastAPI, Power BI, Python, R	Mid	Remote	Whitestad, BB	107786-128719	20-12-2024	Mid	FastAPI, PyTorch, KDB*

Dataset Description



Key Attributes

Company, job title, experience, employment, location, salary.



Experience Levels

Entry, Mid, Senior levels (AI/data).



Work Types

Full-time, Contract, Remote, Internships.



Salary Range

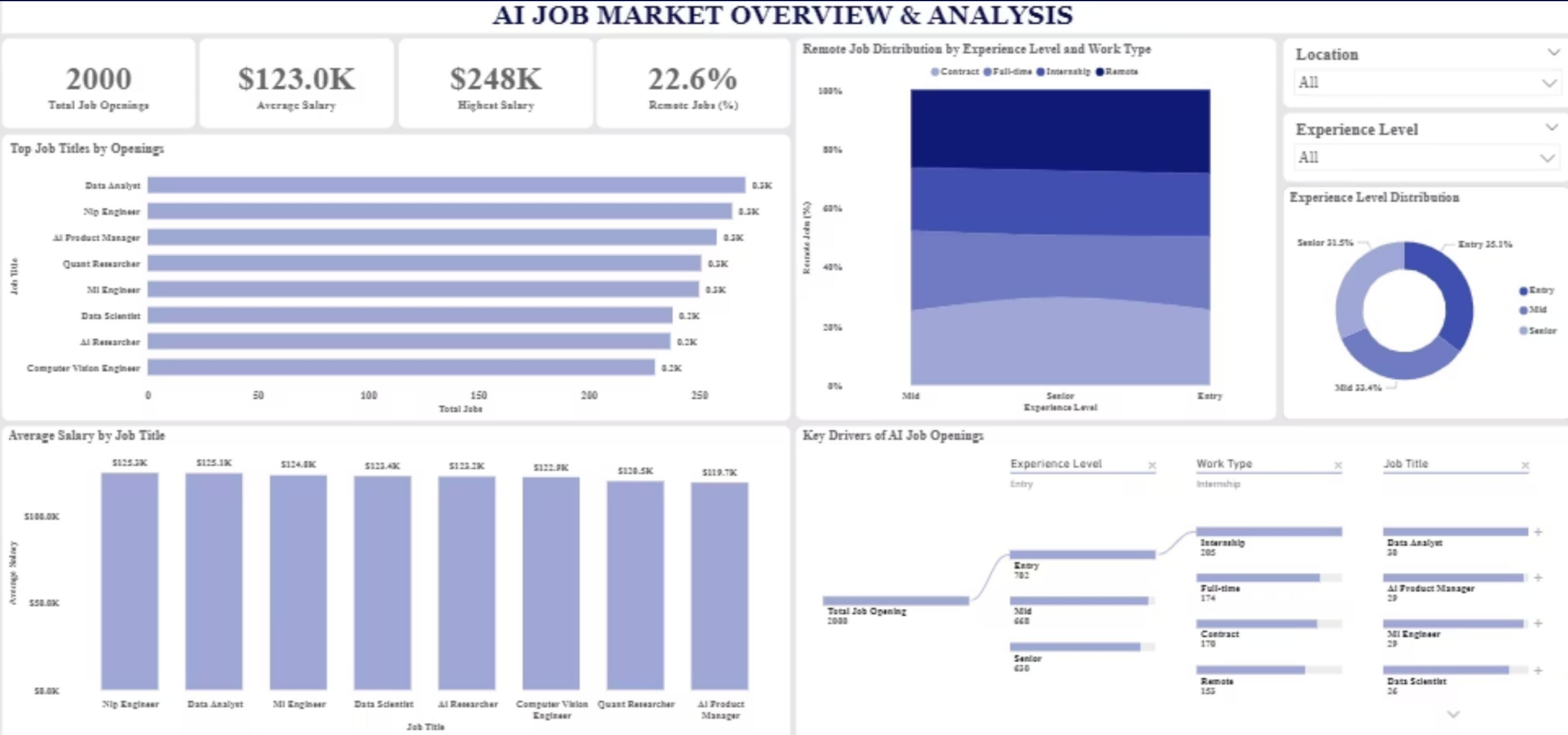
Compensation for all levels; avg. \$123K.



Geographic Coverage

Job openings: multiple locations and remote.

Dashboard – Key Performance Indicators



Dashboard Components & Features



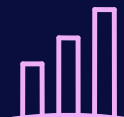
KPI Cards

Key metrics: Total Openings, Avg Salary, Remote.



Top Job Titles

AI job demand by horizontal bar chart.



Average Salary

Average salary by role (vertical bar chart).



Jobs by Company

Job distribution by company (stacked area chart).



Experience Distribution

Experience level proportions (donut chart).



Experience Level Stats

Entry, Mid, Senior stats breakdown.



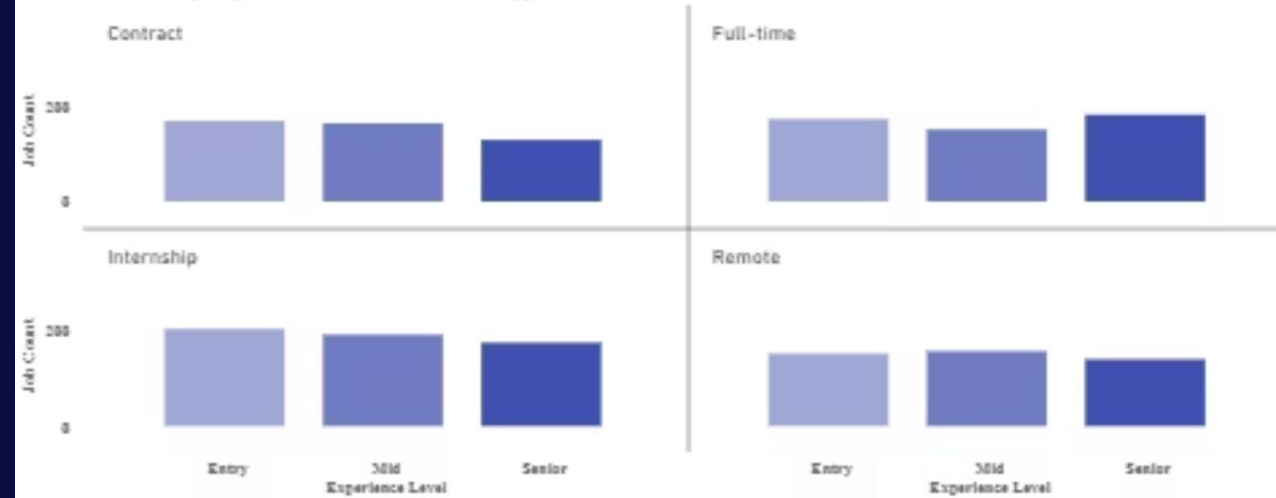
Interactive Slicers

Location & Experience filters.

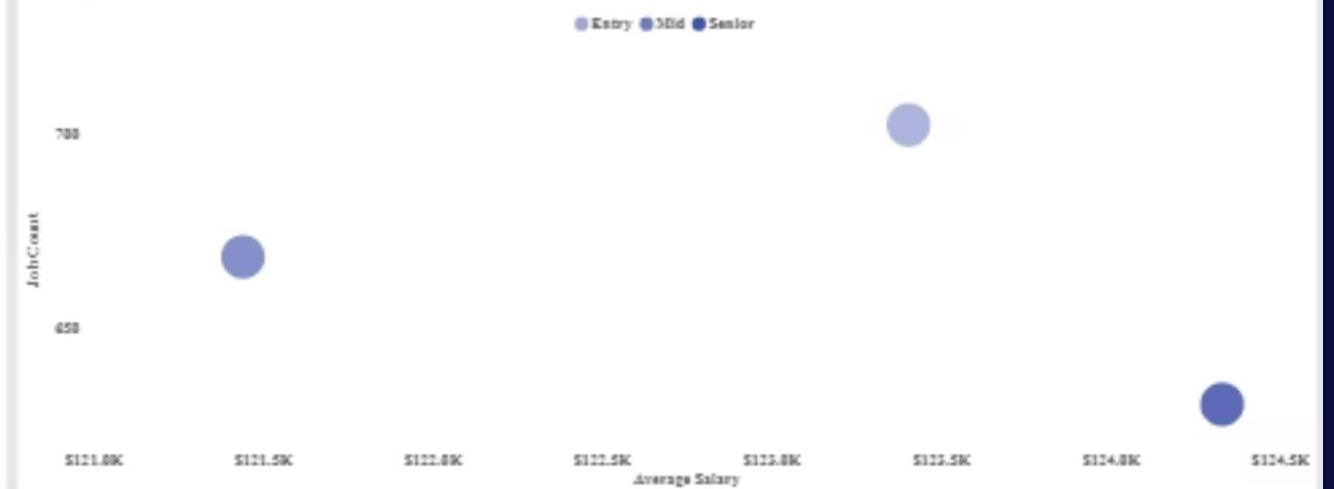
Dashboard – Deeper Dive into Dynamics

AI JOB MARKET DEEP - DIVE ANALYSIS

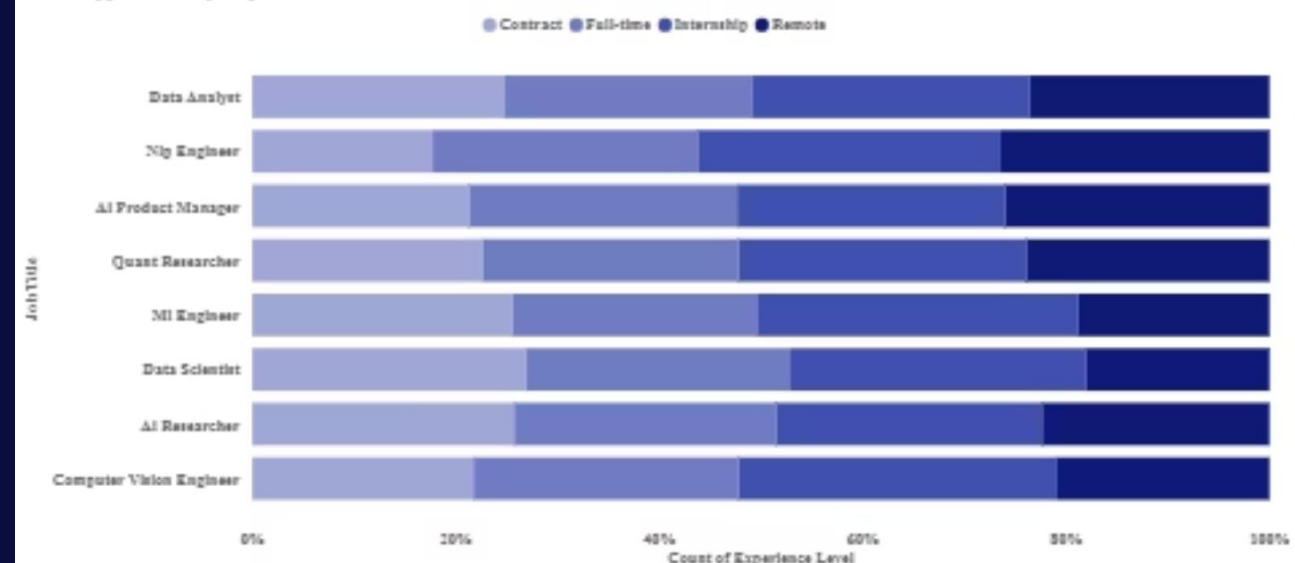
Job Distribution by Experience Level Across Work Types



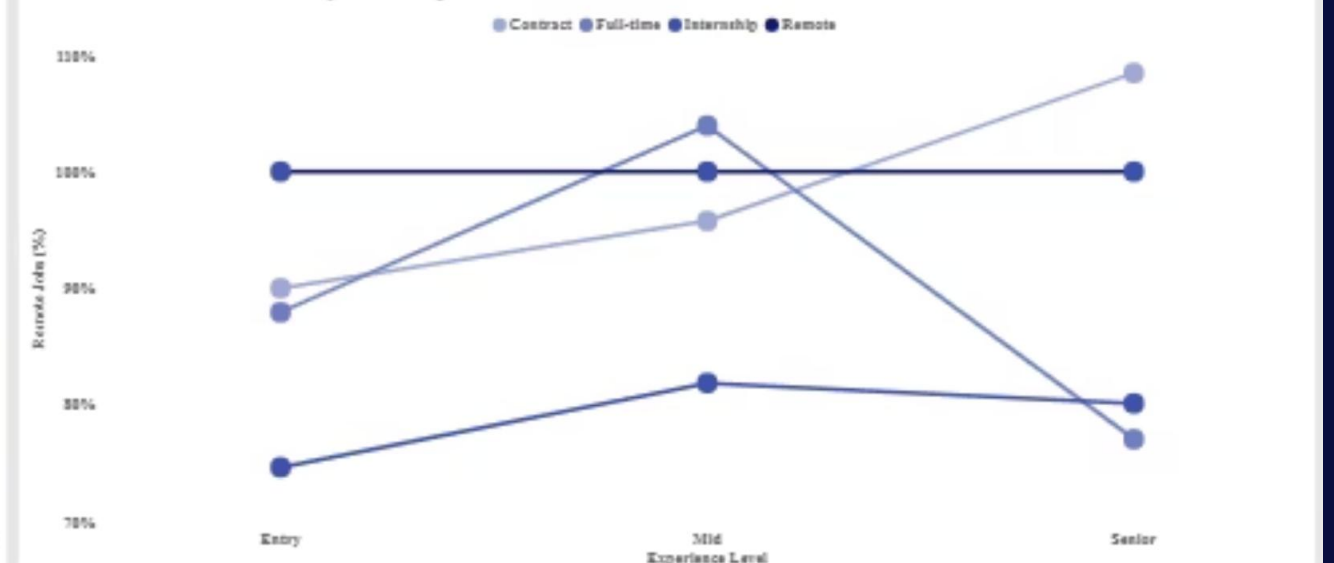
Average Salary and Job Count by Experience Level



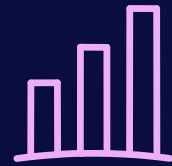
Work Type Share by Experience Level



Trend of Remote Job Availability Across Experience Levels



Advanced Analytics & Filtering



Job Distribution

Counts by experience & work type.



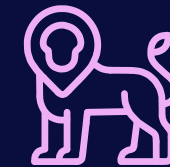
Remote Job Count

Opportunities by experience level.



Top Job Titles

Rankings by experience level.



Remote Job Trend

Across experience levels, over time.

Key Insights : Unlocking the AI Market Landscape



Mid-level Dominance

Mid-level roles dominate AI job openings, indicating strong demand for experienced professionals.



High-Demand Roles

Data Analyst and Machine Learning Engineer are highly sought-after, crucial for data interpretation and AI development.



Senior Salary Premium

Senior-level roles command higher salaries, valuing extensive experience and specialized expertise.



Remote Work & Experience

Remote work increases significantly with experience for seasoned professionals.

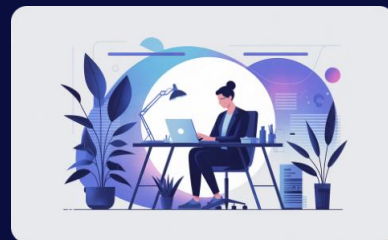
Strategic Recommendations: Navigating Your AI Career



Focus Skill Development: Prioritize mid-level AI skills for abundant opportunities



Upskill for Advanced Roles: Specialize in ML and AI engineering for high demand and competitive salaries

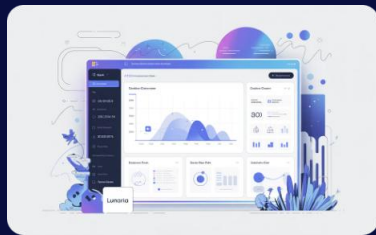


Target Remote-Friendly Roles: Seek remote opportunities as experience grows



Embrace Continuous Learning: Stay current with evolving AI technologies for competitiveness

Tools & Technologies: The Power Behind Our Insights



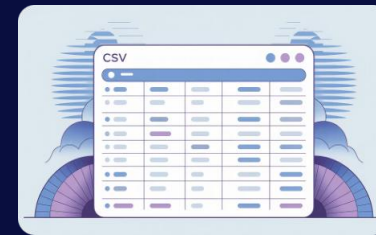
Power BI Desktop

Primary for data visualization and dashboards.



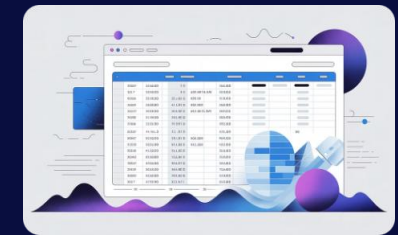
DAX

For advanced calculations and KPIs.



CSV Dataset

Foundational data source.



Microsoft Excel

For initial data cleaning and preparation.

The background is a dark blue gradient. In the center, there is a faint, stylized illustration of two hands shaking in a firm grip. Above the handshake, a lightbulb is depicted with a glowing filament. The entire scene is overlaid with a complex network of thin, light blue lines and dots, resembling a molecular structure or a digital network. The lines and dots are more concentrated around the central elements and fade out towards the edges.

Thank You

I appreciate your time and attention today.